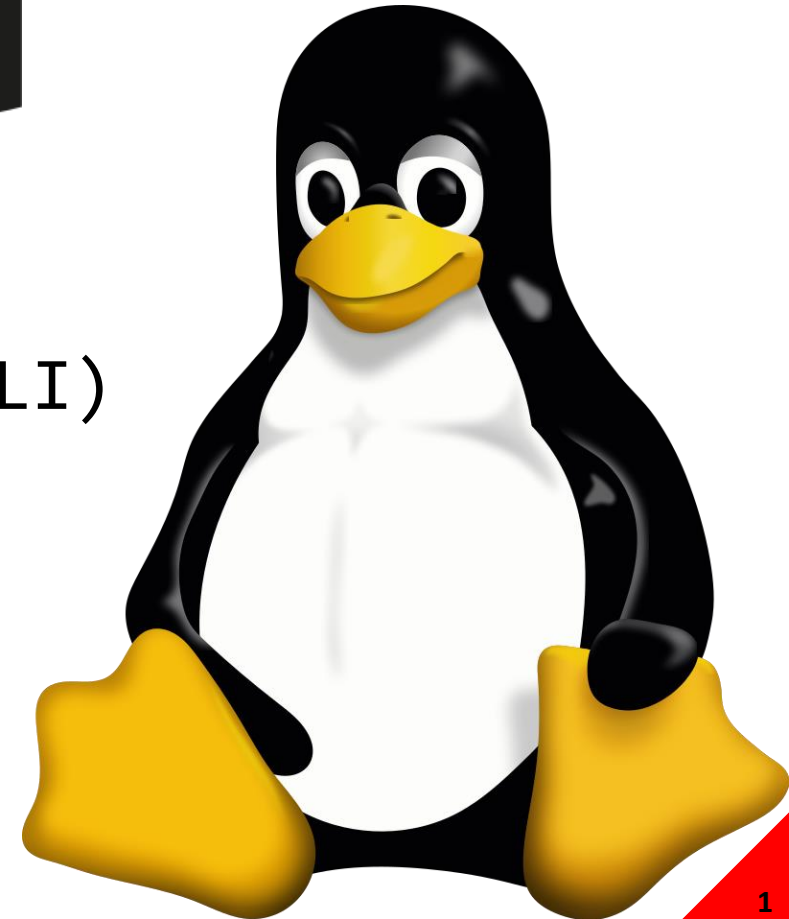




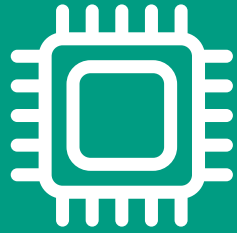
Ubuntu



Command Line Interface (CLI)
&
Scripting



Agenda



Command Line Interface

Trying some commands

+/- 30 minutes



Discussing the assignment

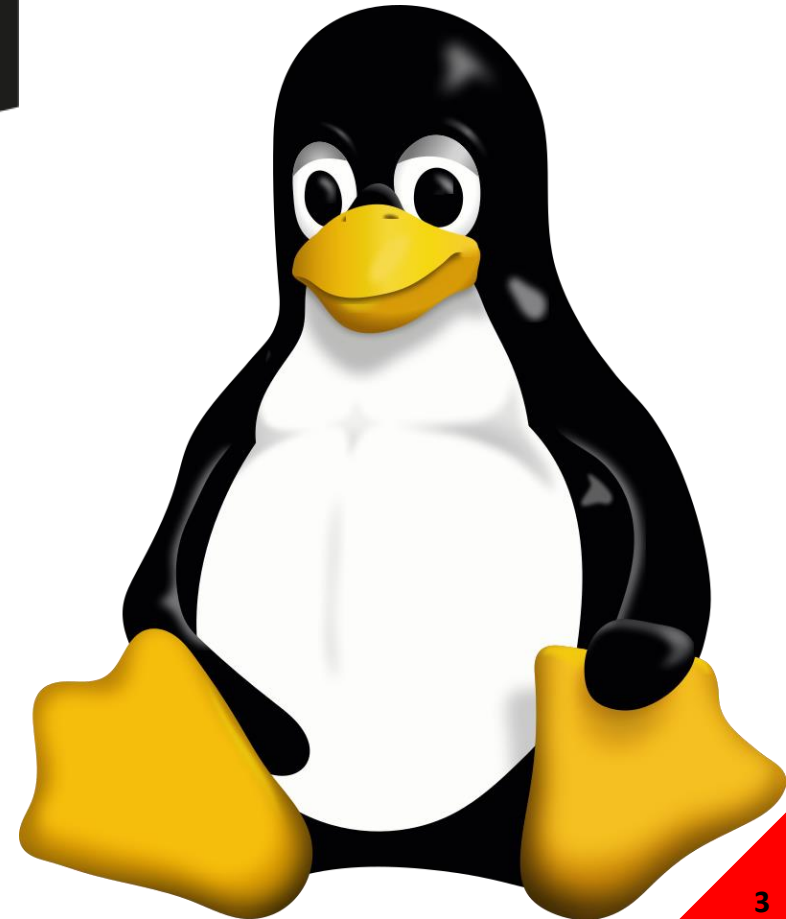
+/- 15 minutes



Ubuntu

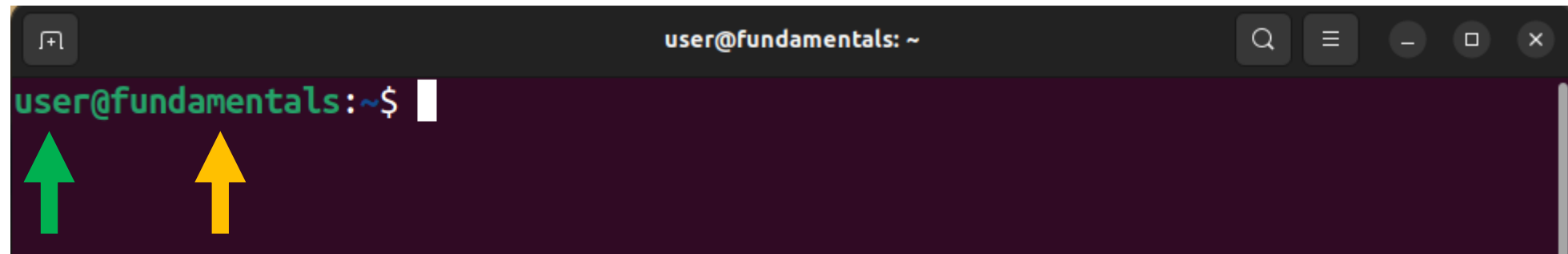
Start your Ubuntu VM in VMware
workstation/fusion

`sudo = super user do`



Linux Terminal – Open the Terminal

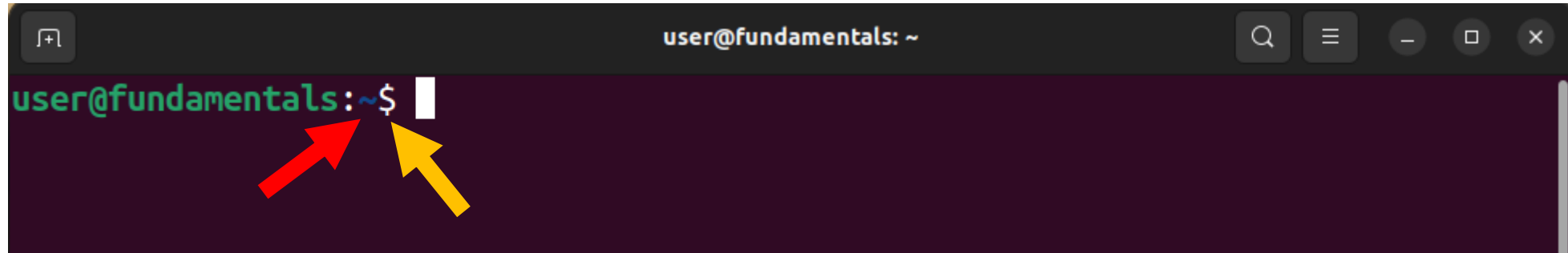
Use the key combination: <ctrl><alt><t>



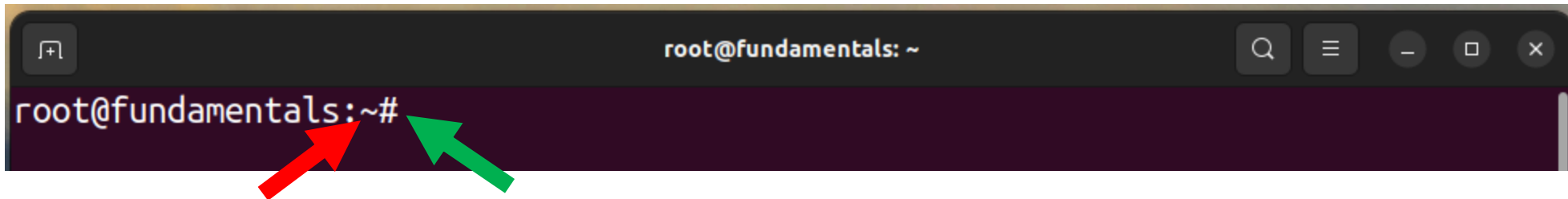
→ • Hostname

→ • Username

Linux Terminal – Meaning of \$, # and ~ sign



A terminal window titled "user@fundamentals: ~" showing the prompt "user@fundamentals:~\$". A red arrow points to the tilde (~) and a yellow arrow points to the dollar sign (\$).



A terminal window titled "root@fundamentals: ~" showing the prompt "root@fundamentals:~#". A red arrow points to the tilde (~) and a green arrow points to the hash sign (#).

- • \$ = normal user
- • # = privileged user
- • ~ = in home directory

Linux Terminal – commands

- • **sudo su –**
 - Become root user
 - • **pwd**
 - print working directory
 - • **exit**
 - logout
 - • **man pwd**
 - What does this command do?
- ! ➤ Be careful when becoming a root user. You can damage your Ubuntu System!

```
manasse@amiga500:~$  
manasse@amiga500:~$  
manasse@amiga500:~$ sudo su –  
root@amiga500:~# pwd  
/root  
root@amiga500:~# exit  
logout  
manasse@amiga500:~$ pwd  
/home/manasse  
manasse@amiga500:~$
```

Linux Terminal – commands: installing software

- • **clear**
 - clear screen
- • **sudo apt update**
 - downloading list of updates
- • **sudo apt upgrade**
 - Installing the upgrades
- • **sudo apt install <package>**
 - Installing a software package

- ? • **Assignment**
 - Find out how to install the **neofetch** package and run it

```
manasse@amiga500:~$ neofetch
      .-/+00ssss00+/- .
    `:+ssssssssssssss++:`
    -+ssssssssssssssyyssss+-
    .0sssssssssssssssdMMMNysss0.
    /ssssssssshdmmNNmyNMMMMhssssss/
    +ssssssssshmydMMMMMMNdddyssssss+
    /ssssssshNMMMyhhyyyhNMMMNhssssss/
    .sssssssdMMMNhssssssshNMMMdssssss.
    +ssshhhyNMMNysssssssssyNMMMyssssss+
    ossyNMMMNyMMhssssssssshmmhssssssso
    ossyNMMMNyMMhssssssssshmmhssssssso
    +ssshhhyNMMNysssssssssyNMMMyssssss+
    .sssssssdMMMNhssssssshNMMMdssssss.
    /ssssssshNMMMyhhyyyhNMMMNhssssss/
    +sssssssdmydMMMMMMNdddyssssss+
    /ssssssshdmmNNmyNMMMMhssssss/
    .0sssssssssssssssdMMMNysss0.
    -+ssssssssssssssyyssss+-
    `:+ssssssssssssss++:`
    .-/+00ssss00+/- .

manasse@amiga500
-----
OS: Ubuntu 20.04.3 LTS x86_64
Host: VMware Virtual Platform None
Kernel: 5.4.0-90-generic
Uptime: 1 hour, 17 mins
Packages: 627 (dpkg), 5 (snap)
Shell: bash 5.0.17
Resolution: preferred
Terminal: /dev/tty1
CPU: Intel i5-10210U (4) @ 2.112GHz
GPU: 00:0f.0 VMware SVGA II Adapter
Memory: 265MiB / 1959MiB
```

Linux Terminal – commands: networking (1)

→ • clear

- clear screen

→ • ip addr (or ip a)

- Show network interfaces
- and ip addresses



```
manasse@amiga500:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 00:0c:29:6a:4f:f3 brd ff:ff:ff:ff:ff:ff
    inet 192.168.121.134/24 brd 192.168.121.255 scope global dynamic ens33
        valid_lft 1054sec preferred_lft 1054sec
    inet6 fe80::20c:29ff:fe6a:4ff3/64 scope link
        valid_lft forever preferred_lft forever
manasse@amiga500:~$ _
```

⇒ • Assignment

- Your VM has one network card, what is the interface name of this network card?
- What is your VM's IP address?
- IP address 127.0.0.1 is a special IP address, explain why.



Linux Terminal – commands: networking (2)

→ • clear

- clear screen

→ • ping google.com

- pinging the site of Google
- Test the internet connection
- Break the ping with the [CTRL]+[C] keys

```
ALICE'S ADVENTURES IN WONDERLAND  
  
Lewis Carroll  
  
THE MILLENNIUM FULCRUM EDITION 3.0
```

⇒ • wget <http://gaia.cs.umass.edu/wireshark-labs/alice.txt>

- Downloading a text file from internet

? • Assignment

- Download the [alice.txt](#) file and try to open it with the **less** command: `less alice.txt`
- Tip: you can quit **less** with the [q] key.

Linux Terminal – commands: files (1)

→ • clear

- clear screen

→ • ls

- List contents of directory

→ • ls -l

- List contents of directory as a long list



• Assignment

- Use the *man ls* command
- How can you list hidden files?
- What does the command *ls -hl* do?

```
ALICE'S ADVENTURES IN WONDERLAND
Lewis Carroll
THE MILLENNIUM FULCRUM EDITION 3.0
```

Linux Terminal – commands: files (2)

⇒ • `head <filename>`

⇒ • `tail <filename>`

```
ALICE'S ADVENTURES IN WONDERLAND  
Lewis Carroll  
THE MILLENNIUM FULCRUM EDITION 3.0
```

? • Assignment

- Use the *head alice.txt* command
- What did this command do?
- Use the command *tail alice.txt*
- What did this command do?

Linux Terminal – commands: files (3)

→ • `cp alice.txt backup.txt`

- Copy file

→ • `mv backup.txt alice2.txt`

- Move file and rename it

→ • `rm alice2.txt`

→ • Remove file `alice2.txt`

? • Assignment

- Use the ***touch myfile.txt*** command
- use the command ***ls -hl***
- What did the ***touch*** command do?

```
ALICE'S ADVENTURES IN WONDERLAND
Lewis Carroll
THE MILLENNIUM FULCRUM EDITION 3.0
```

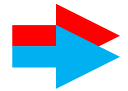
Linux Terminal – commands: files (4)



• nano myfile.txt

- Open file *myfile.txt* in nano editor

```
ALICE'S ADVENTURES IN WONDERLAND  
Lewis Carroll  
  
THE MILLENNIUM FULCRUM EDITION 3.0
```



• Assignment



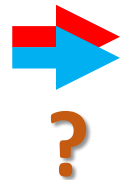
- Add your firstname and lastname to this file.
- Save this file in nano and exit nano
- Use the *more myfile.txt* command to print the contents of the file on the screen
- What does the *cat myfile.txt* command do?

Linux Terminal – commands: files (5)



- mkdir personal

- Make directory personal



- Assignment

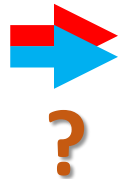
- Move file *myfile.txt* to the directory personal

Linux Terminal – commands: files (6)



- `cd personal`

- Change directory to personal



- Assignment

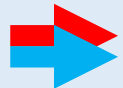
- Go into the directory *personal*
- List the contents of the directory *personal*
- Use the command *cd*
- Use the *pwd* command
- At what location are you now?

Linux Terminal – file permissions



chmod Numeric

Read, Write and eXecute rights on files and directories



For example:

`chmod 754 script.sh`

Will give the owner read, write and execute permissions,

Will give the group read and execute permissions,

Will give the others only read permission,

...on the file `script.sh`

	owner	group	others
read	4	4	4
write	2	2	2
execute	1	1	1
	=====+		
sum	7	5	4



Linux Terminal – file permissions



⇒ chmod Numeric

Read, Write and eXecute rights on files and directories

For example:

⇒ The command **ls -l** will list the contents of the current directory
For each file you can see what kind of permissions they have on them.

```
manasse@amiga500:~$ ls -l
total 156
-rw-rw-r-- 1 manasse manasse 152138 Aug 21  2004 alice.txt
drwxrwxr-x 2 manasse manasse  4096 Nov 25 21:19 personal
manasse@amiga500:~$ _
```

? **d = directory, r = read, w = write, x = execute**

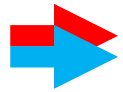
Symbolic	Numeric	Permission
---	0	None
--x	1	Execute
-w-	2	Write
-wx	3	Write + Execute
r--	4	Read
r-x	5	Read + Execute
rw-	6	Read + Write
rwX	7	Read + Write + Execute

Linux Terminal – commands: files (7)



- `cd personal`

- Change directory to personal



- Assignment



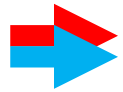
- Go back into the directory *personal*
- List the contents of the directory *personal*
- Change the permissions on the file: *myfile.txt* so that only the owner can:
 - Read and Write the file. The group and the others should have no file permissions.
- Use the `ls -l` command to show that you've successfully changed the file permissions.

Linux Terminal – commands: files (8)



• `cd ~`

- Change directory to user's home directory



• Assignment



- List the contents of the user's home directory
 - Use the command ***ls -hal***
 - What users and groups can you discover in this listing?
 - Use the command ***more .bash_history***
 - Explain what this file logs
-
- The ~ sign means that you are in the home directory

Linux Terminal – commands: files (9)



• Assignment



- Use the command ***cd ~***
- Use the command ***ls -hal***
- Make sure that the ***personal*** directory is present
- Use the command ***man rm*** (*after reading, quit*)
- Use the command ***man rmdir*** (*after reading, quit*)



- Find out how to delete an empty directory.
- Find out how to delete a directory filled with files.
- What command would you use to delete the ***personal*** directory? Why?

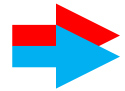
```
manasse@amiga500:~$ ls -l
total 156
-rw-rw-r-- 1 manasse manasse 152138 Aug 21  2004 alice.txt
drwxrwxr-x 2 manasse manasse  4096 Nov 25 21:19 personal
manasse@amiga500:~$ _
```

Linux Terminal – commands



• Assignment

- Use the command **top** or **htop**
- After a while quit to the terminal
- Use the command **man top** or **man htop**
- What does **top** or **htop** do?



```
top - 22:04:42 up 3:40, 1 user, load average: 0.18, 0.18, 0.18
Tasks: 218 total, 1 running, 217 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1959.3 total, 950.1 free, 273.8 used, 735.5 buff/cache
MiB Swap: 3907.0 total, 3907.0 free, 0.0 used, 1516.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
344	root	0	-20	0	0	0	I	0.3	0.0	0:01.05	kworker/2:1H-kblockd
772	root	20	0	310652	7824	6548	S	0.3	0.4	0:34.86	vmtoolsd
1501	root	20	0	0	0	0	D	0.3	0.0	1:45.44	kworker/0:1+usb_hub_wq
1	root	20	0	169472	11404	8344	S	0.0	0.6	0:07.13	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.10	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd



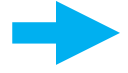
Linux Terminal – commands



• Assignment



- Use the command *cal*
- What does this command do?



- Use the command *man cal*

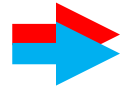


- Find out how to display the year 2030 and run it in the terminal

Own Laptop → Linux VM



• Assignment



- On your own laptop: Start the command prompt on Windows or the terminal on macOS
- Can you ping the Linux VM from your own laptop?

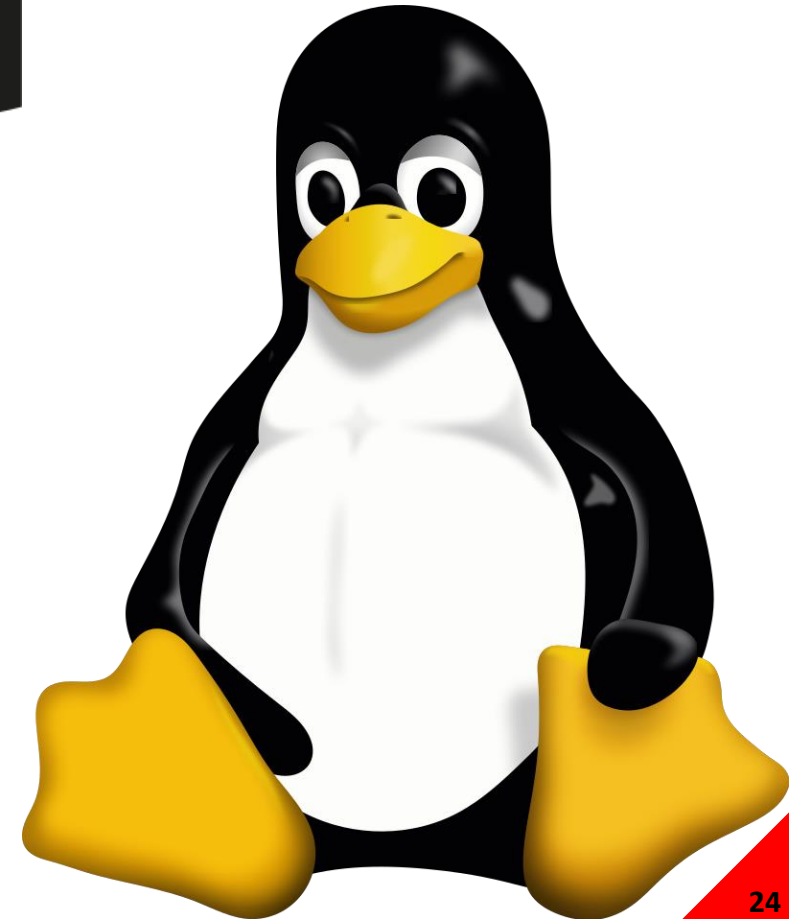
? You need to know the IP address of your Ubuntu VM first...



ubuntu



Scripting



Linux vs Windows commands



Linux cmd	Windows cmd	Meaning
cat < filename>	type < filename>	Show file contents
cd <path>, cd ..	cd <path>, cd ..	Go to directory ; to parent directory
cp <file1> <file2>	copy <file1> <file2>	Copy <file1> to <file2>
ls, ls -l	dir, dir/w, dir/p	List files and directories
mkdir <dirname>	mkdir <dirname>, md <dirname>	Make directory <dirname>
mv <file1> <file2>	rename <file1> <file2>	rename <file1> to <file2>
nano <file1> pwd ps -ax, top, htop rm <file1> rmdir <dir1> rm -r <dir1>	notepad <file1> cd <CTL><ALT> task manager del <file1> rmdir <dir1>, rd <dir1> rmdir /s <dir1>	Edit <file1> Show path of current directory Show current processes Delete <file1>. Delete dir1 if empty Delete dir1 even if full with files
echo hello clear	echo hello cls	Display string on screen. Clear screen

Why scripting?



Suppose we have to issue many commands in succession (and again several times on e.g. different virtual machines).

Disadvantage of doing that all the time:

- Frequent typos
- Takes a lot of effort

Solution:

- Put all the commands together in a file (we call it a script).
- Run that script with just 1 command.

Example:

```
sudo apt update  
sudo apt upgrade  
sudo apt install <package1>  
sudo apt install <package2>
```

BASH (LINUX) SHELL



Listing of File: createfiles.sh

```
#!/bin/bash
```

```
mkdir myfiles
```

```
cd myfiles
```

```
echo "This is the first line" > file1.txt
```

```
cp file1.txt file2.txt
```

```
cp file2.txt file3.txt
```

```
cp file3.txt file4.txt
```

```
cd ..
```

```
ls
```

- Make it executable:
`chmod +x createfiles.sh`
- Then start it:
`./createfiles.sh`





Listing of File: createfiles.sh

```
#!/bin/bash
```

```
mkdir myfiles
```

```
cd myfiles
```

```
echo "This is the first line" > file1.txt
```

```
cp file1.txt file2.txt
```

```
cp file2.txt file3.txt
```

```
cp file3.txt file4.txt
```

```
cd ..
```

```
ls
```

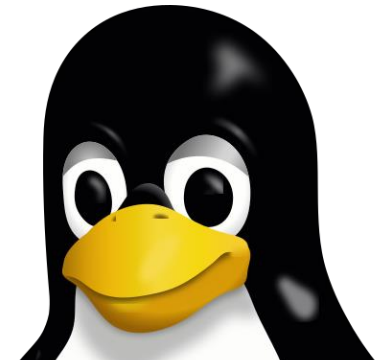
- Make it executable:
`chmod +x createfiles.sh`
- Then start it:
`./createfiles.sh`

Assignment

- Create this script on your Ubuntu VM
 - Make a file called **createfiles.sh** in home directory
 - Edit that file with nano and save it
 - Make the bash script file executable
 - Run it.
- List the contents of the directory myfiles.
- Did the script work?



Command Line Interface (CLI) & Scripting





- In Windows a batch script file uses the **.bat** or **.cmd** extension
 - **.cmd** is the updated version and is used for Windows NT systems (10, 11 etc.)
 - **.bat** files still run on Windows NT systems though, for backwards compatibility.

- Example: [myfirstbatfile.bat](#)

```
@ECHO OFF
mkdir myfiles
cd myfiles
echo hello > file.txt
dir
```

```
MS-DOS Prompt
Auto
C:\>cd scripts
C:\scripts>echo hello > hello.txt
C:\scripts>dir
Volume in drive C is MS-DOS_5
Volume Serial Number is 46F1-81DC
Directory of C:\scripts

.                <DIR>                12-16-23   7:53p .
HELLO            <DIR>                12-16-23   7:53p ..
HELLO            TXT                  8  12-16-23   7:54p hello.txt
                  1 file(s)              8 bytes
                  2 dir(s)             33,310,720 bytes free

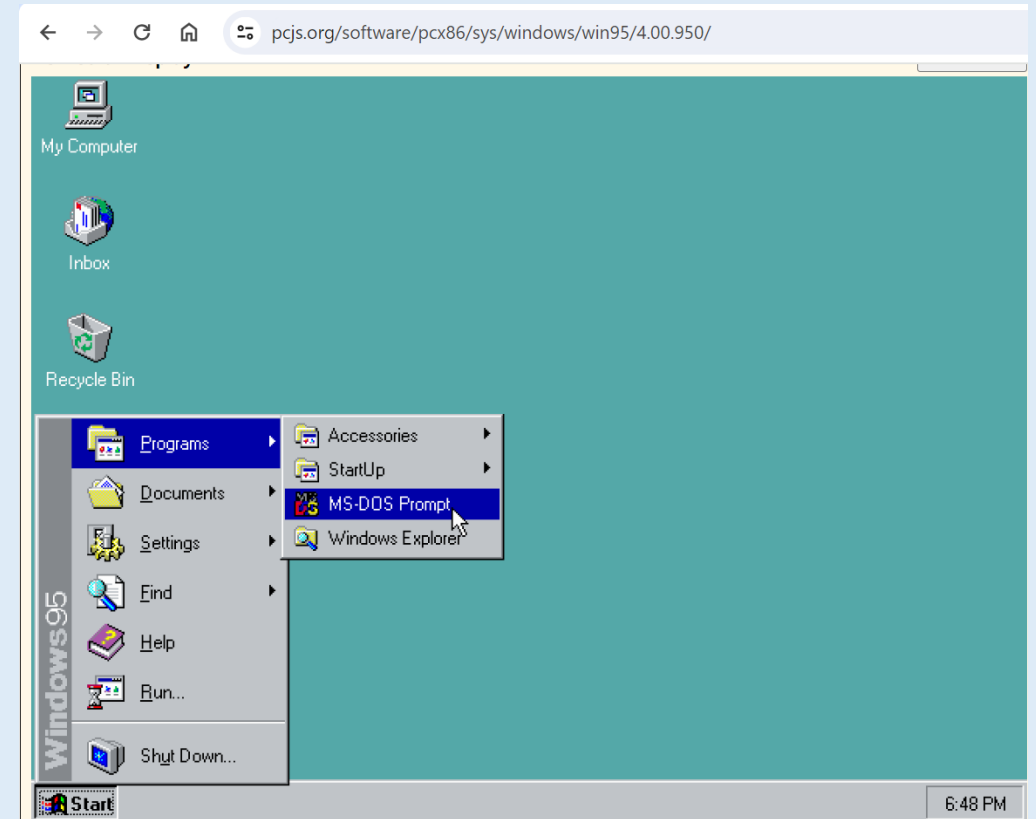
C:\scripts>type hello.txt
hello
C:\scripts>_
```

- Can be started immediately in the command prompt: [myfirstbatfile](#)



Windows 95 VM online

- You can do this Windows scripting assignment with an online virtual machine:
<https://www.pcjs.org/software/pcx86/sys/windows/win95/4.00.950/>
- Scroll down a little bit:
 - You can make the batch scripts in notepad
 - Run them in the MS-DOS Prompt (CLI)
 - Notepad is found under Accessories
- START > Programs > MS-DOS Prompt
- START > Programs > Accessories > Notepad
- But of course, you can also use your VMware VM.



Scripting on Windows

- In Windows create the same script as the Ubuntu Script assignment but translate the Linux bash commands to Windows batch commands. See sheet [25](#) & [27](#) for help.

Listing of File: makefile.bat

```
@ECHO OFF
```

```
MKDIR MYFILES
```

```
.....
```

```
.....
```

```
.....
```

```
.....
```

```
REM FINISH THIS BATCH FILE
```

Assignment

- Create this script on Windows
 - Create a directory **scripts** on the c: drive
 - Make a file called **makefile.bat** in that directory
 - Edit that file with notepad and save it
 - Run it.
- List the contents of the directory scripts.
- Did the script work?

The CLI/Scripting Assignment



Discussing the Assignment

+/- 15 minutes

Have a nice day!

